

On Deterministic LP for Airline Network Capacity Control

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June 24, 2013

1 Introduction

One of the method to optimize network resource is deterministic linear programming problem.

Network optimizer produces shadow prices which serve as input to leg-based optimizer. Optimization on leg level is carried out with EMSR optimization heuristics.

Deficiency is that classic deterministic linear programming could produce inverted values... This is obstacle to apply EMSR optimization method.

Different workarounds proposed which fix the order of values.

In this paper classic deterministic LP is modified to take into account possible inversions of fares for leg-based optimizer. We show that our method produces optimal allocations.

Also uniqueness of solution of LP is studied....

2 Network optimization

Primal LP is

$$p^T x \longrightarrow \max \quad (1)$$

$$Ax \leq c, \quad (2)$$

$$0 \leq x \leq \mu, \quad (3)$$

where

- $p \in \mathbb{R}^n$ is price vector
- $A \in \mathbb{R}^{m \times n}$ is leg-product matrix
- $c \in \mathbb{R}^m$ is capacity vector
- $\mu \in \mathbb{R}^n$ is demand vector

Dual LP to (1),(2),(3) is

$$\lambda_1^T c + \lambda_2^T \mu \longrightarrow \min, \quad (4)$$

$$\lambda_1^T A + \lambda_2^T \geq p^T, \quad (5)$$

$$\lambda_1 \geq 0, \lambda_2 \geq 0, \quad (6)$$

where

- $\lambda_1 \in \mathbb{R}^m$ is shadow price vector of capacities
- $\lambda_2 \in \mathbb{R}^n$ is shadow price vector of demand

Desirability vector is

$$d = p - \lambda_1^T A.$$

Pseudo fare matrix is

$$F = d \otimes \lambda_1^T,$$

where $\otimes$ is tropical product.

Input fares is vector ν satisfying the system of linear equations

$$(\mathcal{A}\mu)^T \nu = \mathcal{A}\mu^T F.$$

Definition 1 Feasible dual solution $\lambda = (\lambda_1, \lambda_2)$ of (4),(5),(6) preserves the order of input fares if

$$(E - E_+) \nu \geq 0.$$

Above E is unity matrix and E_+ is ...

3 Extending dual LP

Extended dual LP is

$$R = \lambda_1^T c + \lambda_2^T \mu \longrightarrow \min, \quad (7)$$

$$\lambda_1^T A + \lambda_2^T \geq p^T, \quad (8)$$

$$(\mathcal{A}\mu)^T \nu = \mathcal{A}\mu^T F, \quad (9)$$

$$(E - E_+) \nu \geq 0, \quad (10)$$

$$\lambda_1 \geq 0, \lambda_2 \geq 0, \nu \geq 0. \quad (11)$$

Theorem 1 Optimal value of (7),(8),(9),(11)

4 Uniqueness of solution for extended dual LP

===== Desirability vector

$$d = p - \lambda_1^T A.$$

Pseudo fare matrix

$$PF = d \otimes \lambda_1^T,$$

where $\otimes$ is tropical product.

Resource-class matrix $\mathcal{A}$.

Input fares for class optimizer is vector ν which is found from the system of linear equations

$$(\mathcal{A}\mu)\nu = \mathcal{A}(\mu^T PF).$$

5 Uniqueness of dual solution

That is needed in order to uniquely identify input fares for leg optimizer. If it is not unique what does it mean?